Training set ($N=3000$) Validation set ($N=1000$) Testing set ($N=1000$) Ideal 1:1 line ($y=x$) ±10% error band

(a) CatBoost

Train: $R^2 = 0.999$, MAPE = 1.2%
Val: $R^2 = 0.997$, MAPE = 2.1%
Test: $R^2 = 0.996$, MAPE = 2.1%

(b) MLP

Train: $R^2 = 0.996$, MAPE = 2.5%
Val: $R^2 = 0.995$, MAPE = 3.0%
Test: $R^2 = 0.995$, MAPE = 2.9%

(c) XGBoost

Train: $R^2 = 0.998$, MAPE = 1.9%
Val: $R^2 = 0.993$, MAPE = 3.3%
Test: $R^2 = 0.993$, MAPE = 3.1%

(d) LightGBM

Train: $R^2 = 0.998$, MAPE = 1.8%
Val: $R^2 = 0.992$, MAPE = 3.7%
Test: $R^2 = 0.990$, MAPE = 3.6%

(e) SVR

Train: $R^2 = 0.992$, MAPE = 4.0%
Val: $R^2 = 0.989$, MAPE = 4.1%
Test: $R^2 = 0.990$, MAPE = 4.1%

(f) RF

Train: $R^2 = 0.995$, MAPE = 2.8%
Val: $R^2 = 0.966$, MAPE = 7.7%
Test: $R^2 = 0.967$, MAPE = 7.6%

(g) DT

Train: $R^2 = 0.981$, MAPE = 5.8%
Val: $R^2 = 0.911$, MAPE = 12.6%
Test: $R^2 = 0.917$, MAPE = 12.2%

(h) Ridge

Train: $R^2 = 0.907$, MAPE = 12.0%
Val: $R^2 = 0.909$, MAPE = 12.1%
Test: $R^2 = 0.894$, MAPE = 12.3%

Predicted T_u (kN·m)
Actual T_u (kN·m)
0 50 100 150 200 250